

SATVIK GARG

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EDUCATION

Simon Fraser University

Bachelor of Computer Science, AI concentration; Full Ride Scholarship

Vancouver, BC

Sep. 2021 – April. 2026

WORK EXPERIENCE

Undergraduate Research Assistant

May 2024 – Aug 2024

Simon Fraser University

Vancouver, BC

- Conducting research on multi-agent reach-avoid games through simulation and real-world implementations
- Designing Mixed Integer Programming and learning-based control logic to maximize number of attackers captured by defenders and scale to higher-dimensional games
- Controlling TurtleBot3 using ROS and Python for real-life simulations, demonstrating method's effectiveness

Software Developer & QA Intern

Jan. 2023 – Aug. 2023

The Boeing Company

Richmond, BC

- Developed E2E/Smoke tests using Java and Selenium for improvement of **45%** in software functionality
- Integrated **Adobe Analytics** on multiple application pages, while creating **JUnit** test cases for accurate tracking
- Cooperated closely with infra. team to support and manage **CI/CD pipeline** and integrate **E2E tests**
- Achieved significant **40%** reduction in reporting time by automating Excel data collection and visualization of key development metrics using **Power BI** and live data from **Azure DevOps**

Ground Station Software Developer

Sep. 2021 – Jan. 2023

SFU Satellite Design Team (SFUSAT)

Vancouver, BC

- Collaborated to build a SATNOGS ground station for SFU Sat's ALEASAT project
- Integrated SATNOGS software by utilizing SDRs, GPredict, and GNU Radio for ground station control
- Programmed the connection between the ground station and the SATNOGS network using a Raspberry PI

PROJECTS

SoundSphere | React Native, NodeJS, PostgreSQL, AWS

Dec 2023 – Dec 2024

- Developed an app that discovers the music taste of others around you by integrating Spotify and Apple Music
- Hosted custom-build REST API on **AWS EC2** instance and the **PostgreSQL** database on **AWS RDS** to ensure scalability and high availability
- Utilized **Redux** and **RTK Query** for efficient state management and data fetching, leading to responsive app performance and user interactions

ConnectPharma | Dart, JS, Git, Firebase

May 2018 – May 2020

- Developed an app from ground-up to connect pharmacists with employers, leveraging Firebase for data storage and **OAuth 2.0/Firestore Rules** for security
- Incorporated features such as job scheduling user-to-user communication/hiring, filter-based candidate search
- Used **Google Cloud APIs** (Places, Distance Matrix) to improve sign-up with auto-completed addresses
- Collaborated with UI/UX designers, pharmacists, and pharmacies to create a user-centric UI in **Figma**

Endless Enemies | C#, Xaml, Unity, Node

May 2018 – May 2020

- Developed a top-down shooter game where the player eliminates enemies surrounding them to earn points
- Used the **A* search algorithm** to control enemy AI's movements, giving them the ability to pursue the player using the smallest path and spawn in tiles where the player may be hiding
- Implemented the Flood Fill algorithm to generate maps and find locations for weapon/item deployment

AWARDS

USRA Scholarship / Undergrad Research Position in Mars Lab

Simon Fraser University Full-Ride Entrance Scholarship with Distinction

Qualification for the VEX Robotics Competition World Championships

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Java, Dart, JavaScript, TypeScript, CSS, HTML, SQL/NoSQL

Frameworks: React, Angular, Maven, PostgreSQL, Node.js, Pytorch, LangChain, Express.js, Numpy